

IoT-based Smart Power Strip and Energy Monitoring

A PROJECT REPORT

submitted by

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled, **IoT based Smart Power Strip with Wi-Fi Control and Energy Monitoring** submitted by

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

This project presents a smart energy management system utilizing two ESP32 microcontrollers operating as a sensor node and an edge node. Each load—comprised of two bulb loads—is monitored using dedicated PZEM modules that measure voltage and current, enabling precise real-time sensing of electrical parameters. The sensor node collects data from the PZEM sensors and publishes it via HTTPS to a cloud broker, while the edge node subscribes to this information, executes control logic based on voltage thresholds (normal, low, and overvoltage), and commands a two-channel relay module to control the connected loads accordingly. This system architecture ensures prioritized load control under varying voltage conditions for energy efficiency and protective operation.

To enhance remote monitoring and data management, the edge node uploads the processed sensor data to the Firebase Realtime Database, facilitating reliable cloud-based data logging and historical energy consumption analysis. Additionally, integration with the Blynk IoT platform provides user-friendly real-time visualization, load control, and alert notifications through the mobile app. The project encapsulates an effective combination of IoT communication protocols, cloud services, and embedded system design to deliver a scalable and responsive smart load monitoring and control solution.

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LIST OF SYMBOLS

S.No	SYMBOL	DEFINITION
1	CT	Current Transformer
2	I	Current (A)
3	V	Voltage (V)
4	PT	Potential Transformer

Chapter 1

INTRODUCTION

The project describes the design of an intelligent energy monitoring and control system with two ESP32 microcontrollers configured as a sensor node and an edge node. Each of the loads, here represented by two bulb loads, is separately monitored through PZEM energy metering modules that can measure voltage and current quite accurately. The sensor node collects this electrical data and publishes over HTTPS to an HTTPS broker, thus allowing IoT-based real-time energy monitoring. The edge node subscribes to the data, processes it by implementing predefined voltage threshold-based control logic, and hence controls a two-channel relay for managing the bulb loads-thereby turning the bulbs ON or OFF based on voltage conditions like normal, low, or overvoltage.

The architecture is designed to adopt cloud-based services for better monitoring and control. The edge node uploads data to the Firebase Realtime Database, whence continuous data logging and historical energy consumption analysis can be performed. The project also incorporates an intuitive mobile app interface for real-time monitoring and remote control of device loads using the Blynk IoT platform. This combination of embedded microcontroller programming, IoT protocols, and cloud storage with mobile app integration offers an effective framework for smart energy management.

This project is motivated by the need for enhancing electrical load management and energy efficiency through the protection of connected devices against voltage anomalies. In this system, real-time automation of voltage and current measurements will enable it to dynamically prioritize loads and take preventive actions. The system is thus scalable and flexible in providing a smart energy monitoring solution with benefits in energy savings, operational safety, and user convenience for both residential and small commercial applications.

1.2 OBJECTIVE:

The major objectives of the project include the development of a smart energy monitoring system on the ESP32 microcontrollers and PZEM sensors for live voltage and current readings to attain effective load management. It is designed to implement voltage-control-based load control using a two-channel relay for regulating two bulb loads to operate at optimal levels under normal, low, and overvoltage conditions in order to safeguard devices and enhance energy efficiency. The system also tries to enable effortless remote monitoring, control, and data logging using the integration of HTTPS communication, Firebase Realtime Database, and Blynk IoT app. This integration allows for increased user interaction, easy access, and efficient energy management. By achieving these objectives, the project aims to deliver a comprehensive, scalable, and easy-to-use IoT-based solution for smart energy monitoring and control suitable for residential or small commercial applications

1.3 ORGANISATION OF THE PROJECT:

The organization of the paper starts with an introduction of the motivation, purpose, and extent of the intelligent energy monitoring and control system. This is followed by a detailed literature review overviewing current IoT-based energy solutions, and gaps identified and laying the premise for the research. The research methodology is then discussed, including the design, hardware, and software elements employed. Later parts of the paper describe system design and architecture, block and functional diagrams, and hardware configuration and software implementation aspects that include sensor integration, communication protocols, and cloud services. System integration is the exhibition of hardware and software module integration with similar functionality. System performance, reliability, and scalability are compared in the results and discussion section. Lastly, the paper wraps up by listing the findings, limitations, and research recommendations to further enhance smart energy management skills.

Chapter 2

LITERATURE REVIEW

2.1 INTRODUCTION:

IoT technologies have now changed the face of energy management, making it possible to monitor, control, and automate electrical loads in real-time. Several IoT-enabled devices are at use, such as smart plugs, sockets, power strips, and energy meters that use microcontrollers like ESP32 in combination with sensors like CT and PT for the improvement of energy efficiency and sustainability. Equipped with cloud platforms like Blynk, these offer remote access, automatic load prioritization, and detailed data analysis. Most of the systems provide solutions for voltage irregularities and energy wastage through user-friendly interfaces with options for connectivity via Wi-Fi and Bluetooth. Despite the innovations, there are certain common shortcomings in most of them, including dependence on internet connectivity, scalability issues, security vulnerabilities, and no capability for offline operation.

Recent research has ranged from low-cost, user-centered smart sockets for simple relay control to the most advanced systems that include environmental sensing and AI-powered load forecasting. The main communication protocols have been centered around Wi-Fi, Bluetooth, and Zigbee, each with an inherent trade-off between range, complexity, and scalability. While some residential models aim to achieve ease of use and notifications, industrial applications focus on precise electrical parameters and optimization of operations. The studies finally underline that integration, increased security, and interoperability will be of great essence in HEMS development. These broad results guide further development toward efficient, adaptive, and secure IoT-based energy solutions.

2.2 REVIEW OF EXISTING STUDIES:

IoT-based solutions for smart sockets, plugs, and energy monitoring systems have been carried out by several researchers, targeting mainly low-cost design, remote control, and automation of households. Ahmad et al. [1] and Joha et al. [4] developed Wi-Fi-based smart sockets using NodeMCU and Blynk for simple switching and overload protection, with simplicity and low cost as their main focus. In fact, Rokonzaman et al. [6] and Mudassar et al. [10] further extended the design by integrating current and

voltage sensing using the ACS712 and ZMPT101B modules, respectively, with cloud connectivity for real-time data logging. However, their systems were only laboratory-scale prototypes; further scaling was not considered.

Bajracharya et al. [2], Ranasighe et al. [3], and El-Khozondar et al. [8] proposed the design and implementation of a smart multiplug and energy monitoring system with either an ESP32 controller or ATmega controller, respectively, and HTTPS and web interfaces for real-time tracking and control. However, these works were more cloud-platform-based and did not support offline data management or edge decision-making. Advanced implementations include those by Ben Dhaou et al. [5], integrating AI with smart meters for energy optimization, and Gozuoglu et al. [14], with hybrid communication-HTTPS, ZigBee, PLC-, and Rao et al. [7] with energy scheduling based on photovoltaic. These systems have been sophisticated and hence restricted to the lab or other controlled environments.

Other works, such as Joya et al. [17] and Rehman et al. [20], implemented Bluetooth and Wi-Fi smart sockets for domestic energy control; however, both systems featured a lack of voltage-based protection and rich multi-load management. In general, the reviewed studies lacked integrated over- and undervoltage protection, dual-node architecture, and comprehensive data logging.

2.3 COMPARISON:

Most of the existing smart plug and socket systems are single-load control-oriented and have limited fault protection, with operation mostly dependent on cloud connectivity. Works such as [2], [6], [9], and [14] involved cloud analytics and HTTPS communication but often lacked edge-level intelligence or local response in case of network failures. Projects [1], [3], and [8] reached low-cost, user-friendly control but remained just prototypes without scalability or comprehensive monitoring.

Only a few works ([5], [7], [15]) demonstrated advanced sensing or AI-based energy analysis, but their complexity and hardware costs were limiting for practical deployment. Most of the protection features were related to overloading and temperature cutoffs ([4], [10], [17]), while not much has been implemented regarding voltage variation detection.

This proposes, compared to those, an IoT-based Smart Power Strip with a dual-node architecture (sensor and edge nodes using ESP32), voltage-based protection through

over/undervoltage control, multi-load management, communication based on HTTPS, and Firebase data logging integrated with the Blynk app. This offers a more reliable, scalable, and energy-aware solution for the integration of real-time monitoring, fault response, and data analytics, thereby eliminating key gaps in previous research.

2.4 RESEARCH GAP:

Despite these advances in IoT-based energy management systems, several research gaps still exist. Prominent among them are security and privacy concerns, with most of the present solutions lacking proper encryption and protective mechanisms that expose devices and data to possible cyber threats. Scalability is limited in many designs since they often support a small number of devices and cannot integrate well with broader HEMS; thus, their applicability in larger or more complex setups is significantly curtailed. Most of the systems also rely heavily on continuous connectivity to the internet, failing to provide any offline fallback modes and hence decreasing the reliability of the system upon network interruptions. The accurate and comprehensive measurement of energy remains an issue, with several implementations relying on load switching with less than full real-time monitoring of electrical parameters such as power factor or energy consumption.

Additionally, advanced control algorithms involving AI or machine learning for predictive energy management and adaptive load control are scant in the literature. Hence, there is limited intelligent automation and proactive energy optimization. User experience problems prevail since most of the systems have limited multi-platform support and lack good voice assistant integration or smooth setup procedures; these will also be barriers to widespread adoption. Closing this list of gaps means improving security, enhancing scalability and offline capabilities, adding sophisticated control strategies, and optimizing user interfaces to further the efficiency and usability of IoT-enabled smart energy management technologies.

Chapter 3

PROPOSED APPROACH

3.1 METHODOLOGY:

This project demands a methodology for the following steps that will ensure that energy monitoring and load control with ESP32 microcontrollers and PZEM sensors are accomplished effectively. Two PZEM modules, which are connected to the Sensor Node ESP32, each sense voltage and current of individual bulb loads. The Sensor Node ESP32 collects and reads electrical parameters from both PZEM modules. It will be programmed to acquire real-time sensor data at regular intervals by applying error handling in order to ensure that the readings are stable and valid.

The second step includes publishing the electrical data collected by the Sensor Node via HTTPS protocol to an HTTPS broker, establishing wireless communication and transferring data to the Edge Node ESP32. The Edge Node subscribes to the data, processes it, and implements load control logic according to voltage levels—deciding on turning the bulbs ON or OFF through the two-channel relay module to act based on normal, low, or overvoltage conditions. Additionally, the Edge Node uploads processed data to the Firebase Realtime Database for cloud storage and logging of historical data. The system also employs the Blynk IoT platform to enable user interaction and remote control, allowing users to monitor real-time electrical parameters and control loads with ease using a mobile app. The entire system is programmed using the Arduino IDE for the integration of all components into one reliable, scalable, and user-friendly smart energy monitoring and management solution. This methodology makes sure that real-time data acquisition, wireless communication, autonomous load management, cloud integration, and user-centric interface design are emphasized.

3.1.1 IOT LEVEL SPECIFICATION DIAGRAM:

Figure 3.1 depicts a system architecture for the IoT-based smart energy monitoring and control project. The hardware involved is a relay, current, and voltage sensor modules, all connected with the ESP32 microcontroller, which collects data and acts on the load. The controller logic of the system runs locally within the ESP32. Sensor data is processed, and relays are actuated based on the real-time voltage and current reading.

HTTPS protocol is used for communication between ESP32 nodes acting as sensor and edge nodes. Efficient transfer of measured electrical parameters from the sensor node to the edge node happens, which in turn performs load management and sends back the response based on voltage conditions. The processed data at the edge node is uploaded to Firebase Realtime Database through cloud APIs for remote storage, analytics, and visualization of the data. User interaction is performed using the Blynk IoT mobile app, which communicates with the cloud platform for real-time monitoring and control. Thus, it is a strong, scalable, and efficient IoT-based smart energy management solution, implemented without incorporating REST communication.

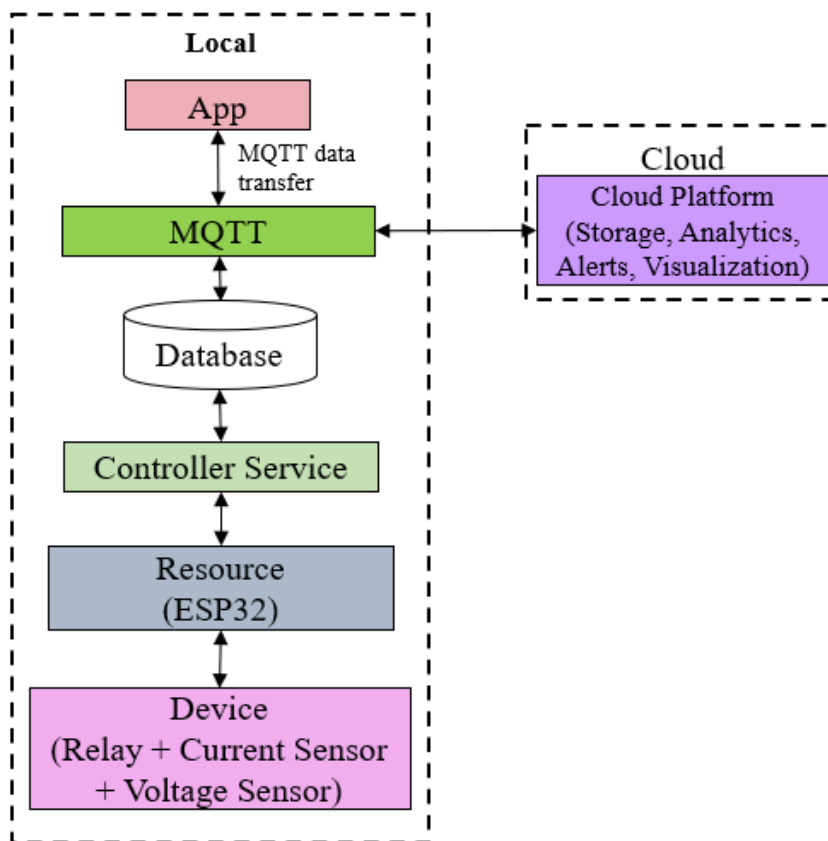


Figure 3.1: IOT level specification diagram

3.1.2 FUNCTIONAL VIEW SPECIFICATION DIAGRAM:

The functional view diagram figure 3.2 outlines a layered structure and data flow for an IoT-based smart energy monitoring and control project. At the device level, it contains ESP32 microcontrollers interfacing with current and voltage sensors and relay modules that collectively provide real-time electrical data and control the

connected loads. These hardware resources are managed by controller services that execute load switching logic and handle local data acquisition. Communication between components is facilitated through the HTTPS protocol, which provides for reliable device-to-node data exchange. Collected data is stored locally and interfaced to higher-level web services for data management, alerts, and firmware updates.

The application layer includes the web app and app server, like Blynk, using the database server, Firebase, for the storage, analysis, and remote visualization of energy data. Critical manageability features such as OTA firmware update capabilities and alert management are included. Security will be provided via authentication and authorization services to provide secured access to the control and monitoring interfaces. The structure shows how real-time monitoring and control, cloud integration, user application access, and the security of a system are managed in order to provide a robust, scalable, and user-friendly smart energy management platform for your project.

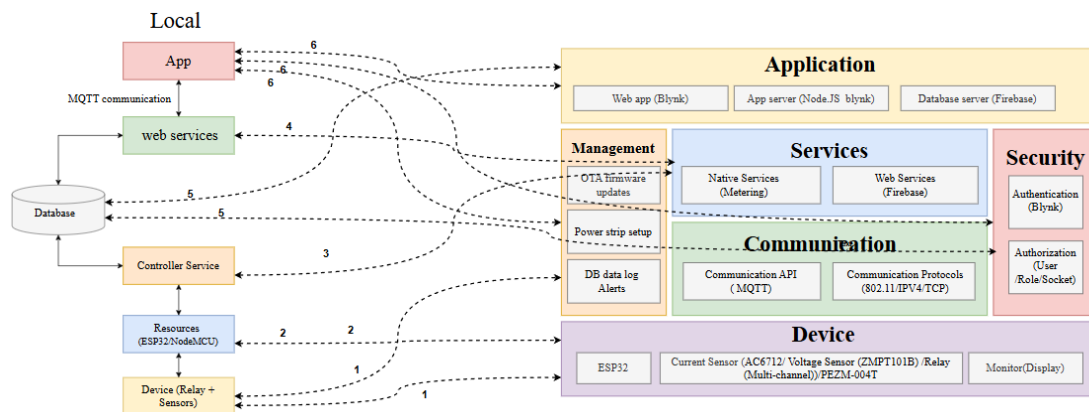


Figure 3.2: Functional view specification diagram

3.1.3 OPERATIONAL VIEW SPECIFICATION DIAGRAM

Figure 3.3 shows the operational view of your smart energy monitoring and control system, highlighting the key functional layers and their roles. To obtain real-time data and control load switching, ESP32 modules communicate with relays, display units, and current and voltage sensors at the device layer. In order to connect device data to higher layers and the cloud, communication depends on HTTPS and common networking protocols. Security features like Blynk-based authentication and authorization coexist with services like energy metering, remote updates, and database

logging to guarantee dependable and secure operations. Smart energy management that is responsive, safe, and easily accessible is provided by application interfaces such as the Blynk web app and app server, which allow users to monitor, control, and analyze system performance through an integrated Firebase cloud database.

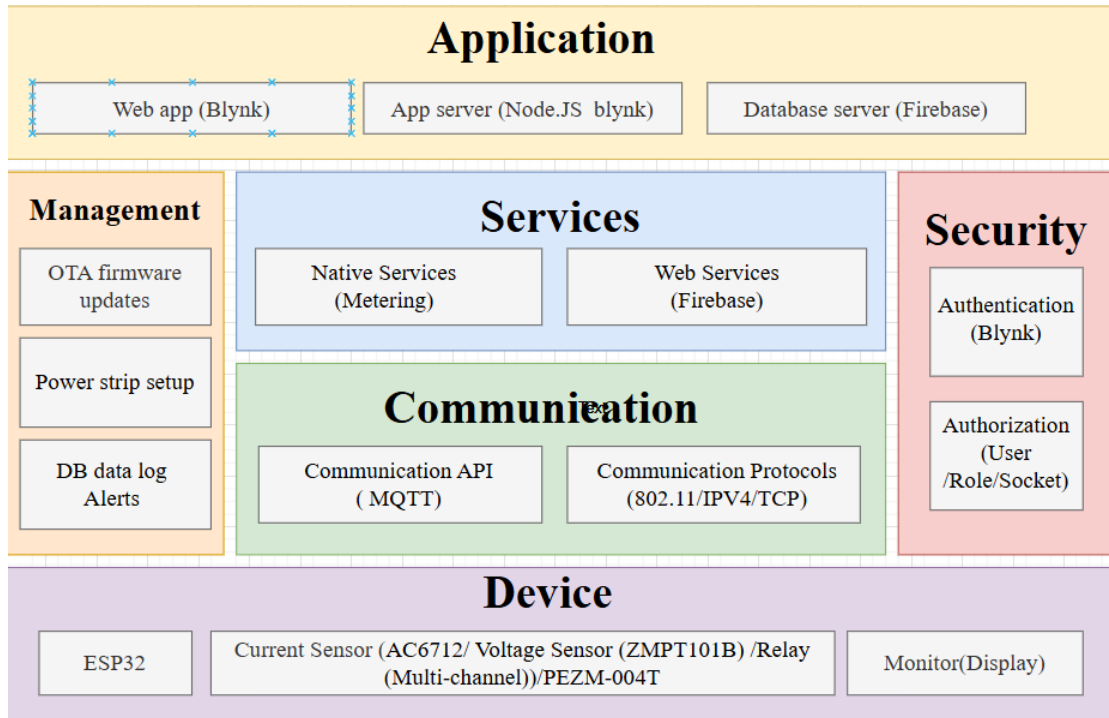


Figure 3.3: Operational View diagram

3.2 PROPOSED FRAMEWORK:

The conceptual architecture of the smart power strip and energy monitoring system is layer-based and is composed of hardware sensors, communication modules, cloud platforms, and user interfaces. Hardware sensors such as current transformers and voltage sensors at the core of the load sense actual electrical parameters in real time. The measurement is also transmitted wirelessly via Wi-Fi modules (e.g., ESP32, ESP8266) to the microcontroller units (MCUs) for processing at the local site. The system uses HTTPS protocol to provide secure message transport between sensor nodes and control units. Data is also streamed to cloud hosts like Firebase for visualizing, processing, and storing when needed, providing a full real-time view. User inputs like load control inputs and energy analytics are managed by mobile and web applications like Blynk for remote monitoring and control. The overall system design is modular, secure, and scalable to accommodate deployment flexibility across various residential

or industrial arrangements and enable energy management, anomaly detection, and prediction analysis features.

The system is also compatible with voltage-based load control algorithms, load priority decision-making, and wastage minimization and energy maximization usage automation procedures. User authentication and data encryption security functionalities safeguard system and user data. Third-party analytics software and energy management integration offers features like high-level reporting, trending analysis, and system maintenance forecasting. The system consolidates low-power sensor devices, low-power communications protocols, safe data storage, and user-friendly interfaces to offer deep, scalable, and secure smart energy management—a maximized process through constant feedback, upgrading, and system optimization for ongoing improvement.

3.2.1 INTEGRATION OF THE SYSTEM DIAGRAM:

Figure 3.4 depicts the unification of the smart power strip and energy monitoring system, describing how the various components communicate with one another in a holistic manner. The Sensor Node ESP32 gathers voltage and current readings from two PZEM modules reporting two different bulb loads. This data is published via HTTPS to an HTTPS broker so that the Edge Node ESP32 can communicate effectively and asynchronously with it. The Edge Node subscribes to sensor data topics and implements voltage-based load control with an operation of a two-channel relay module switching ON/OFF the bulbs based on voltage levels. The Edge Node also publishes real-time sensor data and system status to the Firebase Realtime Database, which supports cloud storage, logging, and analytics. User interaction is enabled through the Blynk IoT smart phone application that provides remote monitoring and control functionalities through exposure to data on the Firebase platform. The integrated framework enables end-to-end communication from sensing to control and on to user application, delivering a scalable, reliable, and interactive IoT-based energy management system.

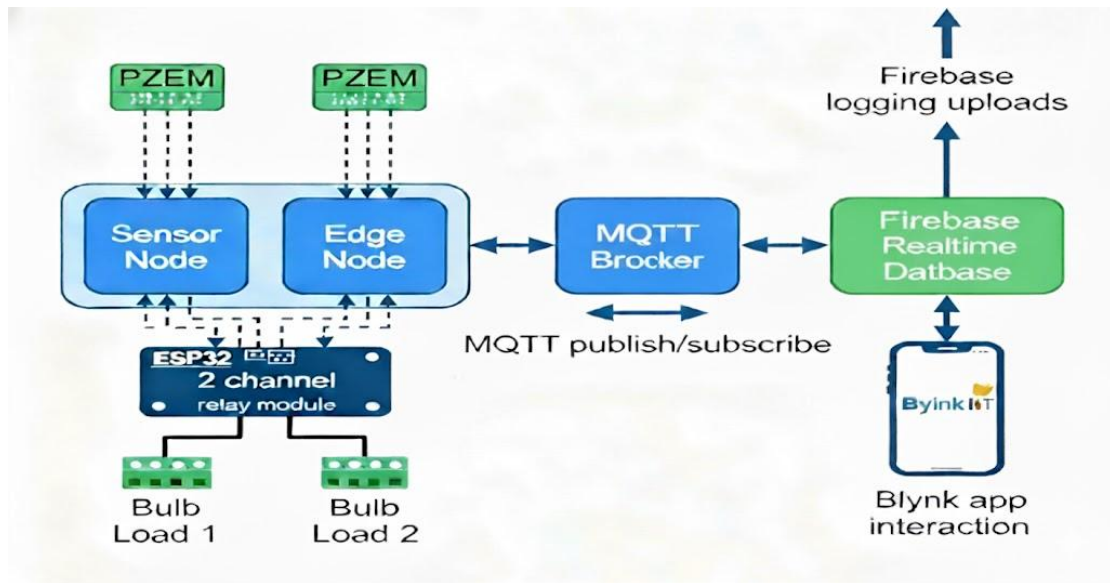


Figure 3.4 : Block diagram

Chapter 4

SYSTEM DESIGN AND ARCHITECTURE

4.1 HARDWARE COMPONENTS

Hardware for the project consists of two ESP32 microcontroller boards used as Sensor and Edge nodes, which supply the processing and communication platform for the project. The loads (light bulbs) are monitored through a PZEM energy meter (e.g., PZEM-004T), which is a dedicated energy meter capable of measuring voltage and current and then transmitting this data to the Sensor Node. These loads are controlled through a two-channel relay module that is interfaced with the Edge Node ESP32 to provide auto-switching of the bulb loads based on voltage logic. Hardware also includes stable AC mains supply power, cabling, and protective devices to provide safe operation and accurate measurement in real-time applications.

4.2 SOFTWARE COMPONENT

The software environment includes various platforms and tools to obtain, process, and access end-to-end data remotely. Firmware for ESP32 nodes is written in the Arduino IDE and corresponding libraries to communicate with PZEM sensors and control relays. Communication between Sensor and Edge nodes takes place using the HTTPS protocol to facilitate efficient publish-subscribe messaging using an HTTPS broker. Data from the Edge Node is streamed to the Firebase Realtime Database for cloud storage and subsequent analysis. For user interaction, the Blynk IoT app is integrated to enable a mobile-based interface for real-time monitoring and remote control of loads. The system also has logic for voltage-based automation, event-driven alerts, and cloud synchronization to increase user convenience and system reliability.

4.3 SENSOR SELECTION WITH END/EDGE NODE & TOOLS/Framework

Node and sensor choice in the system is critical in obtaining accurate and reliable energy monitoring and control. In sensing, PZEM modules are selected because they can deliver accurate voltage, power, and current readings needed for proper energy management. These sensors are equipped with ESP32 microcontrollers, which are employed as end and edge nodes—where the sensor node is utilized for measurement

gathering, and the edge node for data processing, relay management, and interaction with the cloud. This division allows workload partitioning efficiently and system modularity. Software frameworks and tools used are Arduino IDE to write code for the ESP32 nodes, using special libraries to communicate with sensors, HTTPS messaging for fast and fault-tolerant message transmission, and Firebase Realtime Database to store data securely in the cloud. Blynk IoT platform is used depending on ease of integration and support for simple mobile app user interface for real-time control and monitoring. These technologies combined provide a converged platform for scalable, secure, and responsive energy solutions based on IoT.

Chapter 5

IMPLEMENTATION

5.1 HARDWARE SETUP

Figure 5.1 depicts the hardware setup, where there are two ESP32 microcontroller boards acting as a Sensor Node and an Edge Node. Each of the bulb loads is monitored by dedicated PZEM modules, which can measure voltage and current. The normal connection would be to power the PZEM-004T modules at 3.3V or 5V - whatever is supported - connect PZEM TX to ESP32 RX2 (GPIO 16) and PZEM RX to ESP32 TX2 (GPIO 17) for serial communication, and establish a common ground between modules. AC side: connect live (L) and neutral (N) from the AC supply securely to the PZEM's AC input terminals to measure load parameters with high accuracy. It connects the Edge Node ESP32 to a two-channel relay module, where the relay IN1 and IN2 pins are connected to appropriate digital GPIO pins in the microcontroller to control two bulb loads. The common and normally open terminals of the relay are interconnected with AC load wiring for safe isolation and actuation. All low-voltage (signal and power) wiring is strictly kept segregated from AC mains wiring as a best safety practice. Special care in making solid electrical connections and proper isolation by using electrical tape or appropriate protective enclosures is taken, especially when arranging the CT coils and terminal blocks for the PZEM modules. This complete hardware assembly ensures secure power distribution, reliable sensor communication, and safe, responsive load switching for effective system operation.

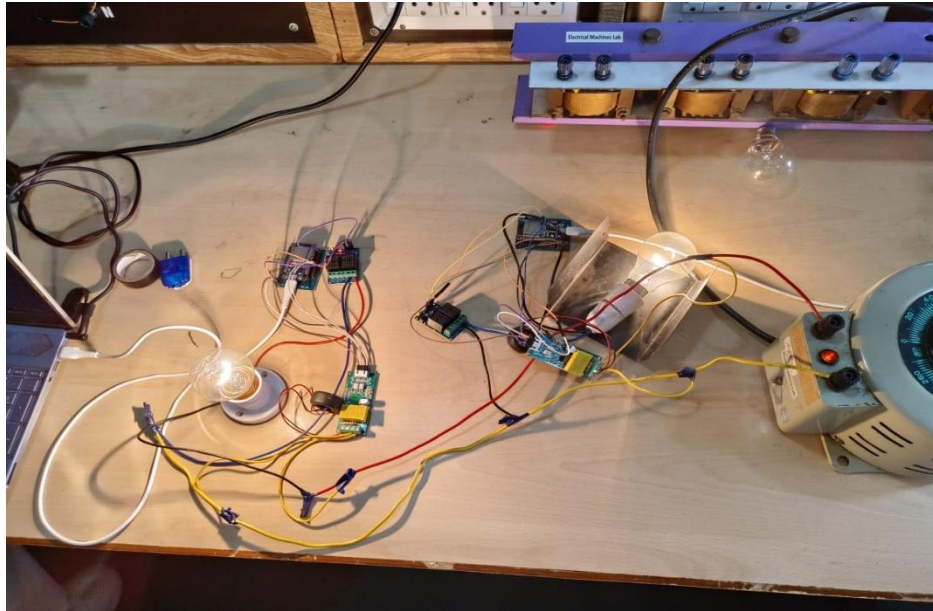


Figure 5.1 Hardware setup

5.2 SOFTWARE DEVELOPMENT

System integration in this project deals with integrating hardware and software modules into a unified, complete IoT-based energy monitoring and control system. It starts with the integration of ESP32-based Sensor Node and Edge Node using the HTTPS publish/subscribe communication model. The Sensor Node will read real-time voltage and current data from the PZEM modules and publish to the designated HTTPS topics, whereas the Edge Node subscribes for the same to receive live updates. Processing this information, including voltage-based control logic, is done by the Edge Node, which actuates a two-channel relay module for efficient management of the bulb loads.

This is followed by local communication and bridging to the cloud layer, which connects the Edge Node to the Firebase Realtime Database through the Firebase ESP Client library for transmitting sensor data and the status of the system. This allows for the logging and visualization of data, including remote storage, for its continuous tracking and analysis performance. On the user interface side, the Blynk IoT mobile application interacts with Firebase through an intuitive dashboard that offers real-time monitoring, manual control, and system notifications. In this respect, the integration enables software-hardware modules to work in a harmonious manner, ensuring seamless data flow, responsive control, and reliable cloud interaction for a complete end-to-end smart energy management framework.

5.3 SYSTEM INTEGRATION

The system integration phase in this project focuses on combining the hardware and software modules to create a unified and fully functional IoT-based energy monitoring and control system. Integration begins by linking the ESP32-based Sensor Node and Edge Node through the HTTPS publish/subscribe communication model. The Sensor Node reads real-time voltage and current data from the PZEM modules and publishes the data to designated HTTPS topics, while the Edge Node subscribes to those topics to receive live updates. The Edge Node processes this information, applies voltage-based control logic, and actuates the two-channel relay module to manage the bulb loads efficiently.

Following local communication, the system bridges to the cloud layer by connecting the Edge Node to the Firebase Realtime Database using the Firebase ESP Client library for transmitting sensor data and system status. This enables data logging, visualization, and remote storage for continuous performance tracking and analysis. On the user interface side, the Blynk IoT mobile application interacts with Firebase, providing an intuitive dashboard for real-time monitoring, manual control, and system notifications. The integration ensures synchronized operation between all hardware and software modules, facilitating seamless data flow, responsive control, and reliable cloud interaction for a complete end-to-end smart energy management framework.

Chapter 6

RESULT AND DISCUSSION

The Blynk IoT dashboard snapshots depict the exact implementation of the voltage-dependent load control logic programmed by the system. In Figure 5.1 Blynk Dashboard – Undervoltage State, Voltage_1 and Voltage_2 are below 150 V, therefore both relays remain ON (marker '1') but the readings of current and power are zero, which means that both bulb loads are disconnected in response to undervoltage protection. This safeguards appliances during adverse supply conditions.

In Figure 5.2 Blynk Dashboard – Single Load Operation, observed voltage on both channels increases into the ≥ 150 V and < 200 V region. The system logic activates only one relay ('Relay_1' is '0') and provides power to one bulb load. This is reflected in non-zero values of Current_1 and Power_1 since Relay_2 is in the off state and the associated load is turned off (current and power are zero). This operation confirms selective load operation for medium supply voltages, which offers maximum energy usage and reserves partial availability.

Figure 5.3 Blynk Dashboard – Overvoltage State shows both Voltage_1 and Voltage_2 to be more than 240 V, tripping the two relays (indicators reset to '1'), and both loads are de-energized as a safety precaution in case of overvoltage. The prompt nature of relay triggering and recovery of the current and power to zero in the two channels confirm the robustness of the system to protective response to voltage abnormality.

These results confirm that the ESP32-dual-node system, with real-time HTTPS communication and cloud-linked control, successfully enacts voltage-based switching scenarios: no loads for undervoltage (< 150 V), selective enablement of loads from 150 to 200 V, full enablement from 200 to 240 V, and auto-tripping for overvoltage (≥ 240 V). The user interface by Blynk accurately replicates all transitions, providing assurance of robust, remote, and auto load management.

Figure 5.4 "Blynk Dashboard – Overcurrent Condition" shows the system reporting an abnormal condition of overcurrent, low Voltage_1 and Voltage_2, and suddenly high Current_1 and Power_1, indicating an abnormal load or fault on Channel 1. The dashboard records current and power with the relays ON, which proves that the system can show hazardous conditions and may take protective measures in the form of warnings or de-energizing the faulty circuit to avoid damage/danger. This proves the

efficacy of IoT-based monitoring for voltage and current safety. Firebase is indispensable in this project because of its role in efficient data management and real-time synchronization between hardware and user interface. In the implemented system, sensor readings comprising voltage, current, and power values uploaded by the Edge Node are sent to the Firebase Realtime Database. This cloud-based platform enables efficient data logging, retrieval, and remote access to allow users to monitor parameters and control loads from any location via the Blynk IoT app. Justification for Firebase: it is highly scalable, has low latency, and natively supports JSON data structures, making it optimal for IoT applications needing live updates. Besides this, it will ensure secured data handling through authenticated connections and perform bidirectional synchronization, including when connectivity fluctuates, ensuring that every update of sensors and actions of relays immediately reflect in all devices. Thus, Firebase acts as the core data backbone in achieving reliable visualization, control, and persistence of energy consumption tracking in this smart energy monitoring system.

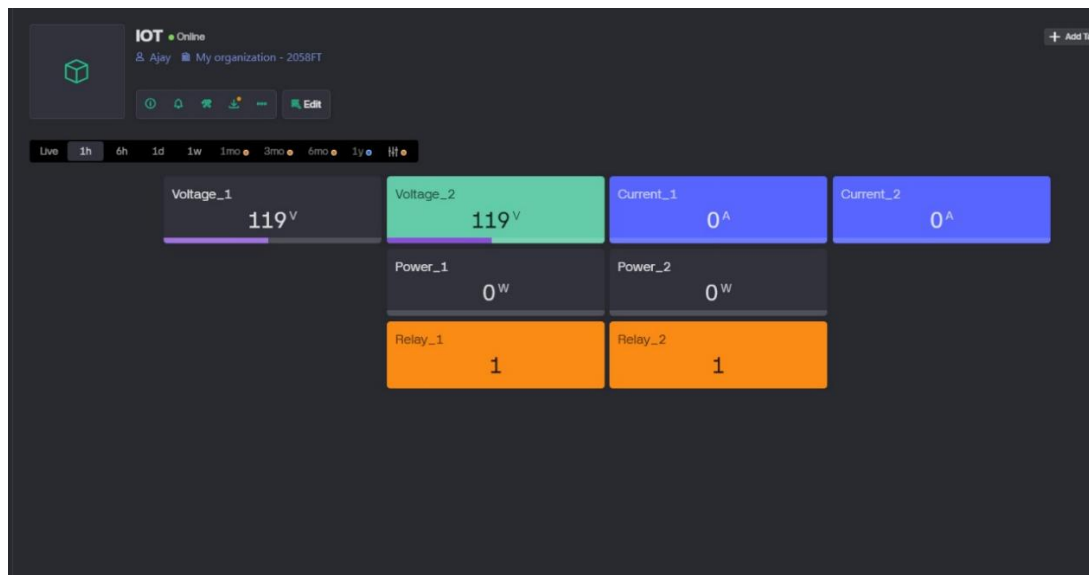


Figure 6.1: Blynk Dashboard – Undervoltage State

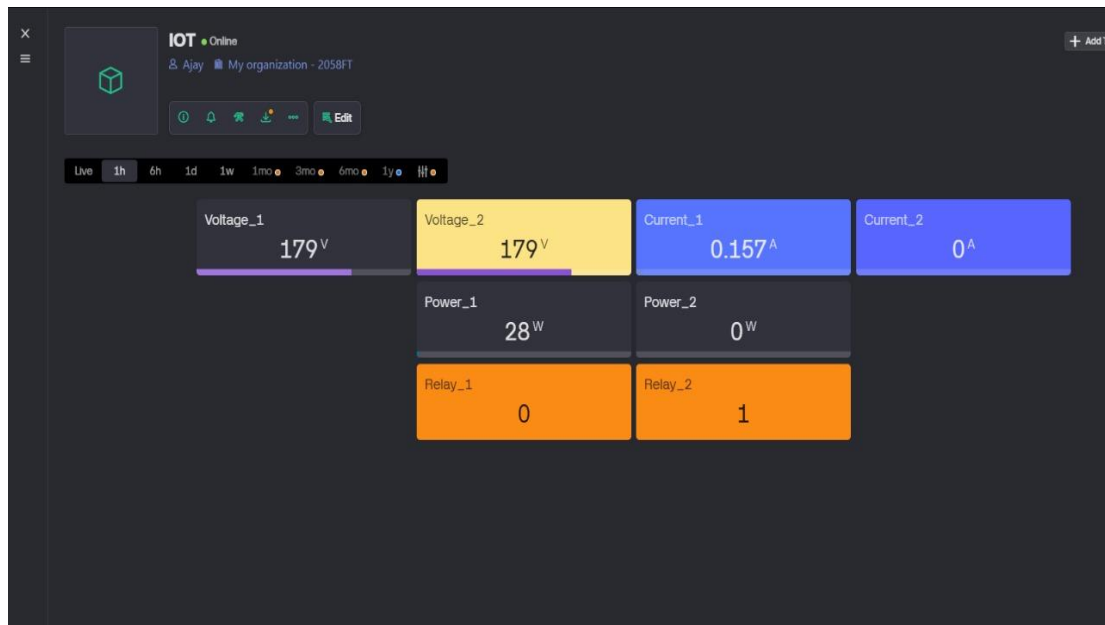


Figure 6.2: Blynk Dashboard – Single Load Operation

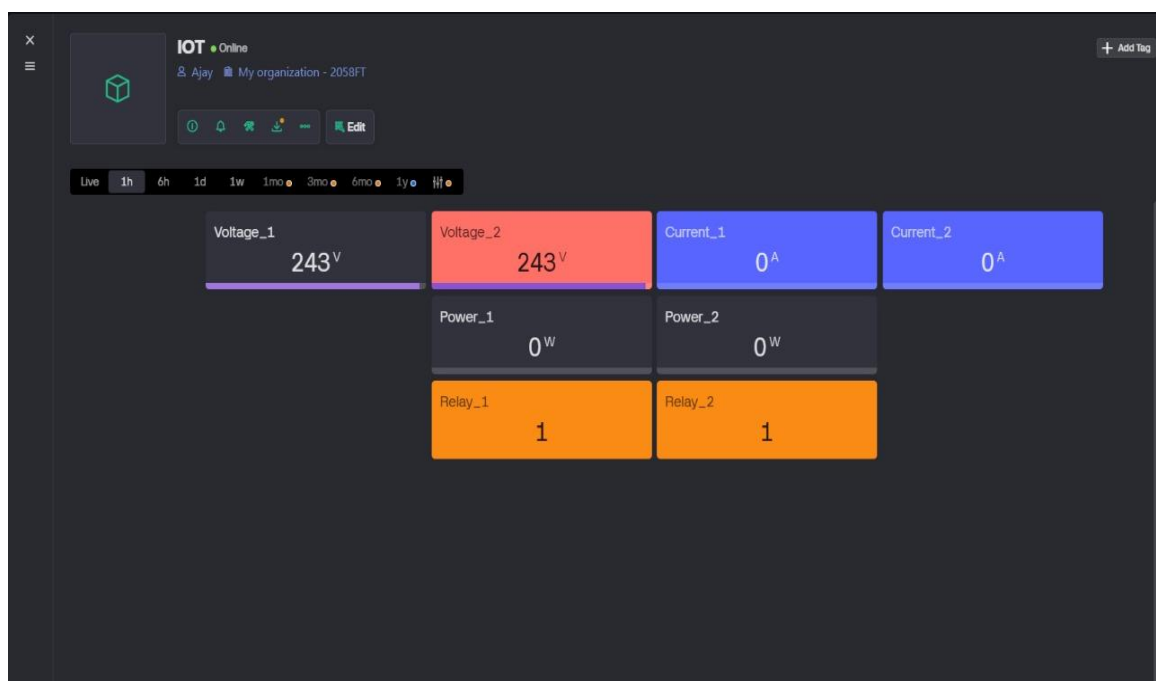


Figure 6.3: Blynk Dashboard – Overvoltage State

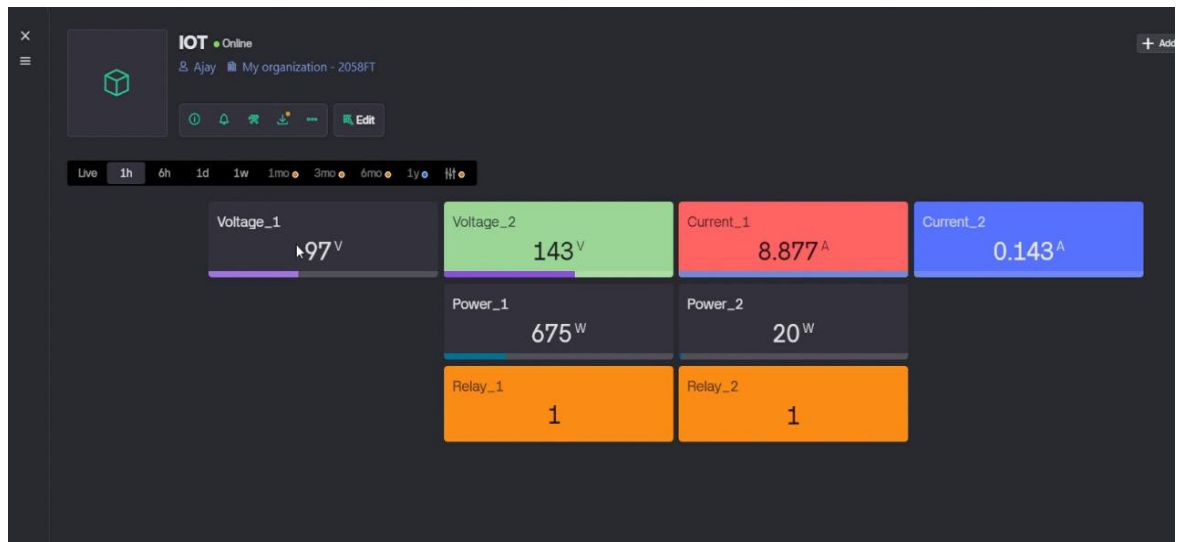


Figure 6.4: Blynk Dashboard – Overcurrent State

Chapter 7

CONCLUSION AND FUTURE WORK

The suggested smart power strip and energy monitor system simply illustrates how to connect ESP32 microcontrollers, PZEM sensors, HTTPS communication, cloud-based data logging through Firebase, and easy management using the Blynk IoT platform. The system measures and regulates voltage and current for single electrical loads in real-time. The system supports optimal and automatic voltage-mode load control. By facilitating smooth communication among cloud software and hardware, it provides users with unobtrusive monitoring, remote access, and analytical capabilities that enhance productivity, conserve power, and secure devices.

In addition to its direct advantages, this IoT-based solution encourages smarter, more efficient energy management. It provides users and facility managers with full insight into usage patterns and insight information for intelligent decision-making. The inclusion of mobile apps and cloud services increases convenience and availability. It facilitates rapid response to abnormal situations and allows for easier deployment of data-driven waste-reducing measures. Modular and scalable system design allows for residential, business, and small industrial environment support and adds value to different environments.

For further growth, the system can be enhanced with predictive maintenance, load forecast, and user behaviour and environment-based automation machine learning algorithms. Make it more sensor and control device compatible, enhance security, introduce offline fallback, and implement automatic optimization algorithms to increase its reliability further. Develop multi-platform dashboards and add voice assistant capabilities to enrich user interaction, making the platform even more universal for future smart energy use cases.

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